Econ 812 Recitation 1 Notes

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1 Notes on Probelm Set 1

1.1 Question 1

Show that

$$\lim_{\sigma \to 1} \frac{c^{1-\sigma} - 1}{1-\sigma} = \ln(c)$$

Should be able to use L'Hospital's rule

$$\lim_{x \to c} \frac{f(x)}{g(x)} = \lim_{x \to c} \frac{f'(x)}{g'(x)} \tag{1}$$

paired with the exponent rule

$$a^b = e^{b \ln(a)} \tag{2}$$

1.2 Question 2

Show that the intertemporal elasticity of substitution is

$$\sum_{t=0}^{\infty} \beta^t \frac{c^{1-\sigma} - 1}{1-\sigma} = \sigma^{-1} \tag{3}$$

The intertemporal elasticity of substitution is

$$-\frac{\partial \ln(c_{t+1}/c_t)}{\partial \ln(u'(c_{t+1})/u'(c_t))} \tag{4}$$

1.3 Question 3

Show that if the production function $F(x)$ is constant returns to scale and input and output markets are competitive, then profit is zero.

Constant returns to scale means

$$F(\alpha K, \alpha L) = \alpha F(K, L) \tag{5}$$

that is, the marginal product of production is constant.(Niepelt section 3.1.2 might also be helpful here.)

1.4 Question 4

Show that factor shares of a Cobb-Douglas production function are constant when input and output markets are competitive. Cobb-Douglas

$$F(x_1, x_2, \ldots, x_n) = x_1^{\alpha_1} x_2^{\alpha_2} \ldots x_n^{\alpha_n} \tag{6}$$

with $\alpha_i > 0$ and $\sum_{i=1}^{n} \alpha_i = 1$
Factor share is defined as

$$\frac{q_i x_i}{pF(x)} \tag{7}$$

1.5 Question 5

No hint here

2 Example with CRRA

Constant relative risk aversion is a type of utility function given as:

$$u(c) = \frac{c^{1-\sigma} - 1}{1 - \sigma} \tag{8}$$

Where σ is the relative risk aversion. If $\sigma > 0$, then $u(c)$ is concave (Figure 1). If $\sigma < 0$ then $u(c)$ is convex (Figure 2). If $\sigma = 0$ then $u(c) = c$ (Figure 3) and if $\sigma = 1$ then $u(c) = \ln(c)$.

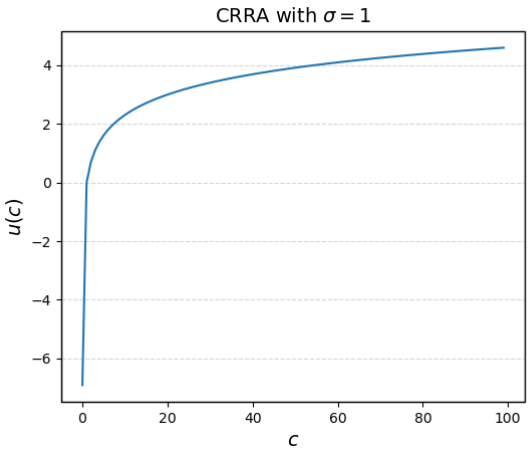


Figure 1: Example risk averse behavior

From class, we had that

$$\max u(c_0) + \beta u(c_1) \tag{9}$$

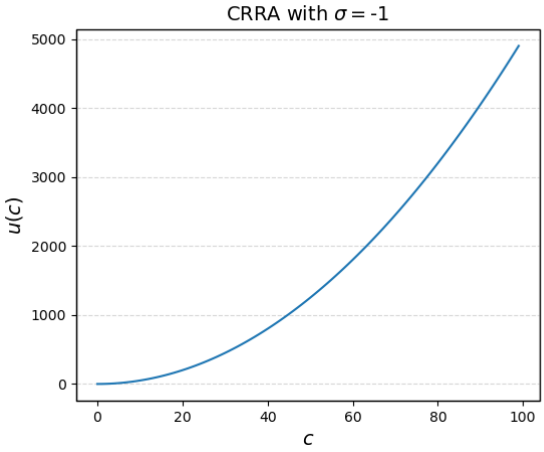


Figure 2: Example of risk seeking behavior

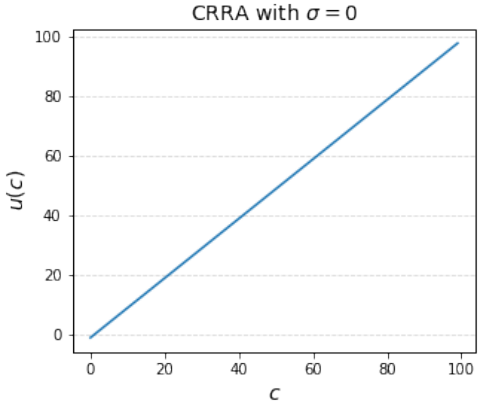


Figure 3: Example of neither risk seeking nor risk averse

subject to

$$c_0 + A_1 = RA_0 + y_0 \tag{10}$$

$$c_1 + A_2 = RA_1 + y_1 \tag{11}$$

Where the left hand side is the consumption / savings decision, and the right hand side is the exogenous income and predetermiend state variable

Set up Lagrangian as

$$\mathscr{L} = u(c_0) + \beta u(c_1) + \lambda_0[y_0 + RA_0 - c_0 - A_1] + \lambda_1\beta[y_1 + RA_1 - c_1 - A_2] - \beta\mu_1[-A_2 - 0] \tag{12}$$

Taking FOC's / CS conditions leads to

$$c_0 : c_0^{-\sigma} = \lambda_0 \tag{13}$$

$$c_1 : \beta c_1^{-\sigma} = \beta \lambda_1 \tag{14}$$

$$A_1 : \lambda_0 = \beta \lambda_1 R \tag{15}$$

$$A_2 : \beta \lambda_1 = \beta \mu_1 \tag{16}$$

$$\lambda_0 : c_0 + A_1 = RA_0 + y_0 \tag{17}$$

$$\lambda_1 : c_1 + A_2 = RA_1 + y_1 \tag{18}$$

$$\mu_1 : -A_2 = 0 \tag{19}$$

Solve for the Euler Equation to be

$$c_0^{-\sigma} = \beta R c_1^{-\sigma} \tag{20}$$

Combine this Euler Equation with the budget constraints to get

$$(RA_0 + y_0 - A_1)^{-\sigma} = \beta R (RA_1 + y_1)^{-\sigma} \tag{21}$$

Simplify to solve for A_1 to get

$$A_1 = \frac{(\beta R)^{1/\sigma}(RA_0 + y_0) - y_1}{(\beta R)^{1/\sigma} + R} \tag{22}$$

Substitute this back into the budget constraints to get c_0 and c_1 to get

$$c_0 = RA_0 + y_0 - A_1 \tag{23}$$

$$c_1 = RA_1 + y_1 \tag{24}$$